Telecommunications

Practical VoIP Implementation

FreePBX®
let freedom ring™

1. Introduction

Sirs Systems is a small not-for-profit corporation supporting start-up enterprises through remote and virtual working techniques. The organization seeks to minimise operating costs by transitioning towards a regionally distributed workforce model employing Voice over Internet Protocol (VoIP) technologies.

The goal of this practical implementation was to create and configure a prototype VoIP system capable of supporting distant workers utilising FreePBX and Asterisk technologies. The installation aims to provide essential business communication capabilities including SIP extensions, voicemail, Interactive Voice Response (IVR), Ring Groups, conferencing facilities and VoIP security procedures.

The system was implemented utilising a Debian 12 virtual machine running FreePBX within VMware Workstation 17 Pro. Multiple SIP softphones were established across Windows and Android devices to simulate geographically spread consumers operating across remote locations.

2. Network Design

2.1 Overview of Network Architecture

The VoIP system was based around a centralised FreePBX server installed on a Debian 12 virtual machine. SIP softphones operating on Windows and Android devices were linked to the PBX server via the local network environment.

The implemented solution provides:

- Remote SIP extensions
- Voicemail services
- IVR functionality
- Ring Groups
- Audio conferencing
- Blacklisting
- VoIP security configuration
- Secure SIP transport support
- Music on Hold

2.2 Network Diagram

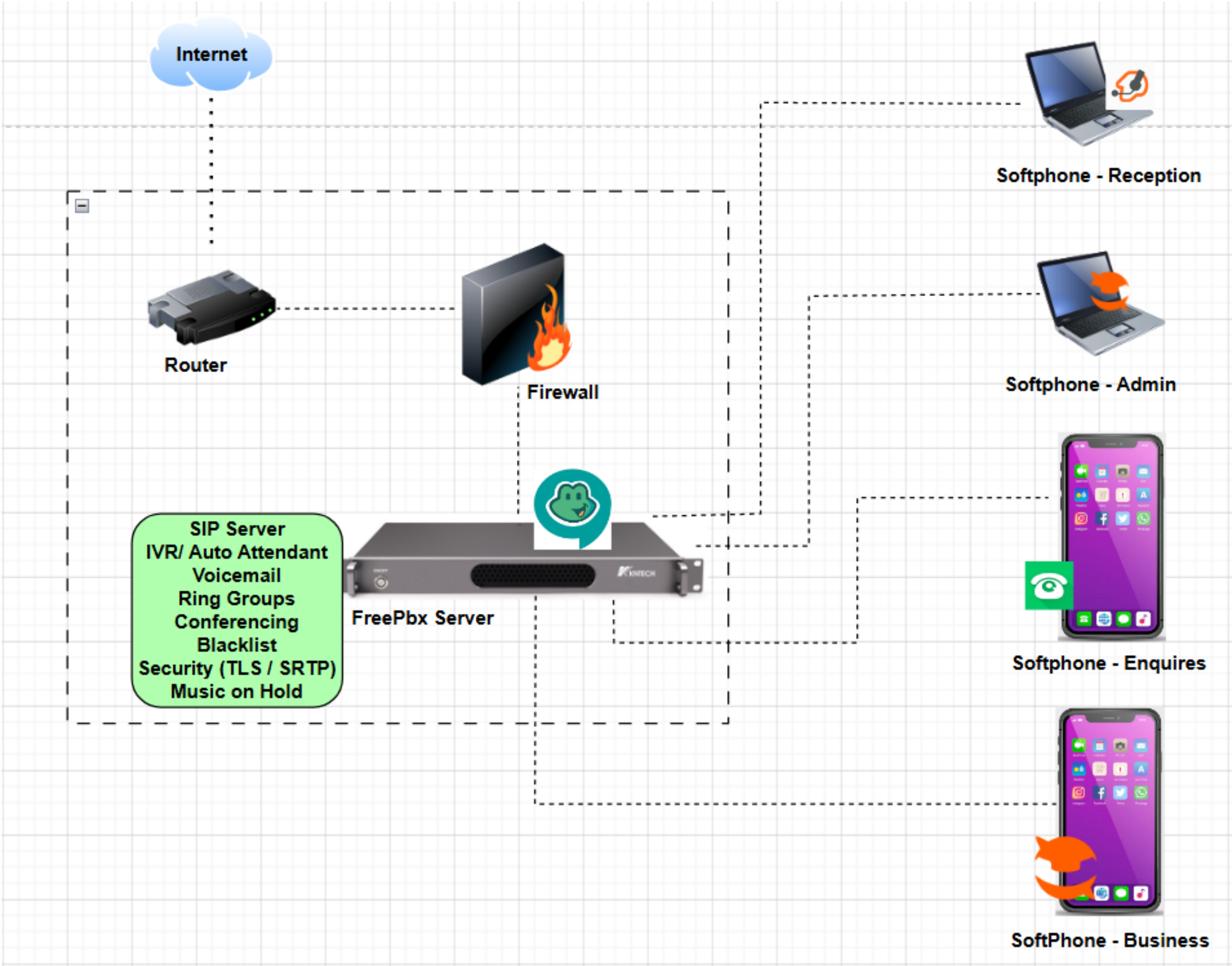
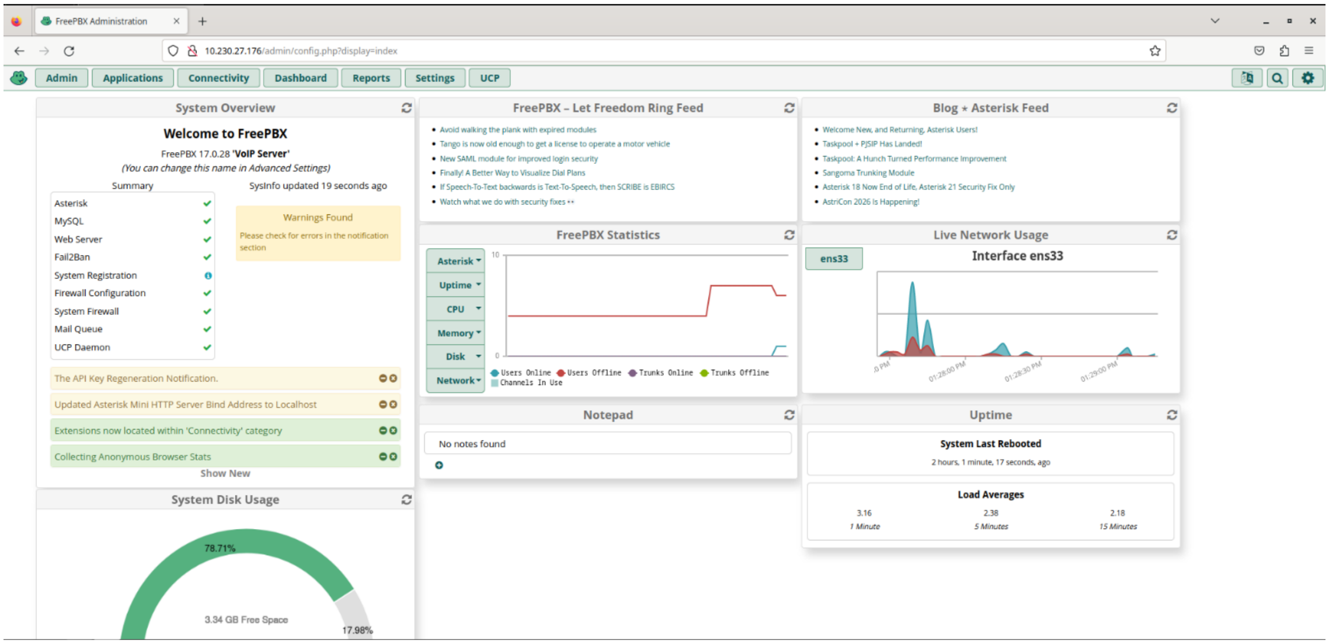


Figure 1: Top-level VoIP network architecture for the Sirs Systems PBX implementation.

2.3 PBX Server Environment

Component	Configuration
Hypervisor	VMware Workstation 17 Pro
Operating System	Debian 12
PBX Platform	FreePBX
Server IP Address	10.230.27.176



3. Dial Plan Design

3.1 Extension Plan

Extension Number	Purpose
2000	Reception
2100	Administration
2200	New Business Enquiries
2300	Business Support
6000	Staff Ring Group
6100	Support Ring Group
7000	Main IVR
8000	Conference Room
*97	Voicemail

3.2 Dial Plan Justification

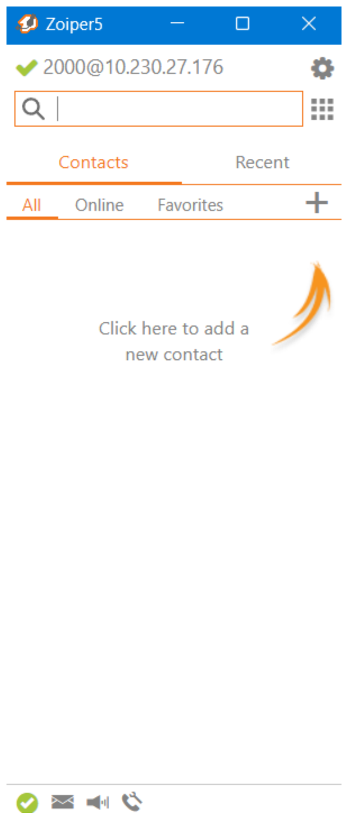
The dial plan was built utilising a systematic numbering method to increase scalability, usability and organisational clarity. The 2000 range was designated for individual SIP extensions, whereas the 6000 range was allocated for Ring Groups. The 7000 range was designated to IVR services and the 8000 range was reserved for conference facilities.

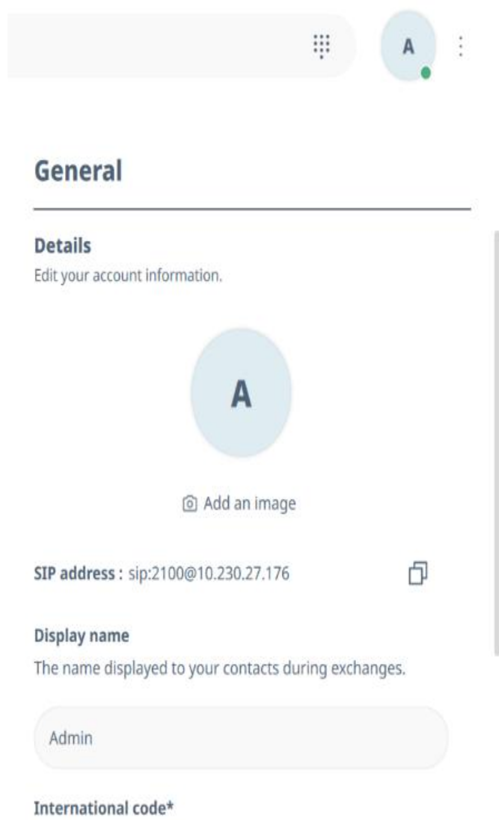
This framework facilitates future growth while retaining logical separation between consumers, services and organisational units.

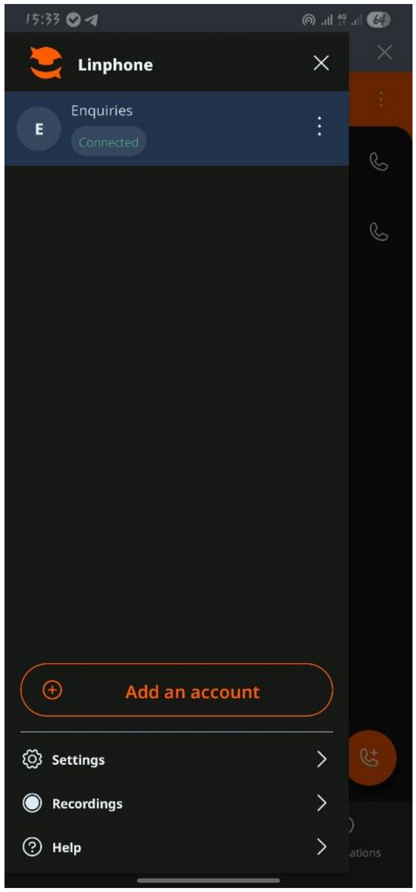
4. Level 1 System Implementation

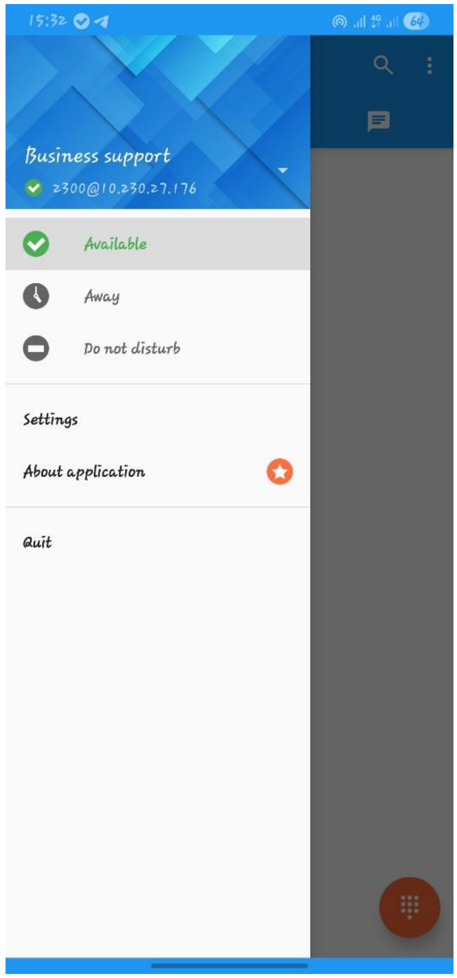
4.1 SIP Extension Configuration

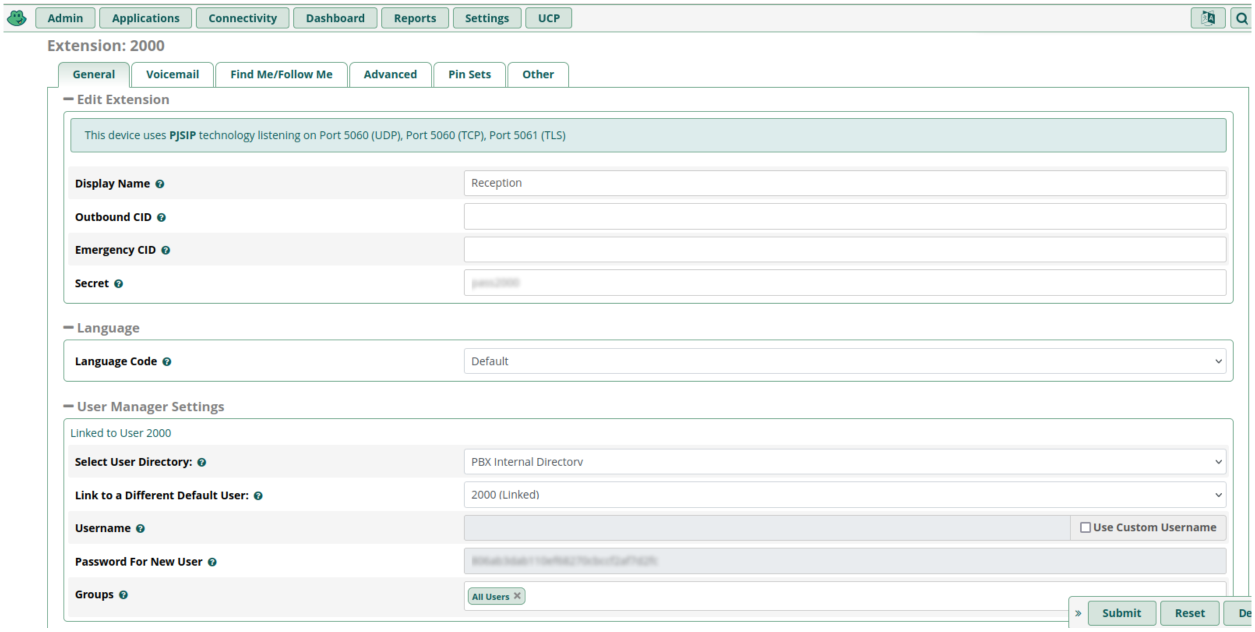
Four SIP extensions were configured within the FreePBX environment to simulate geographically spread consumers.

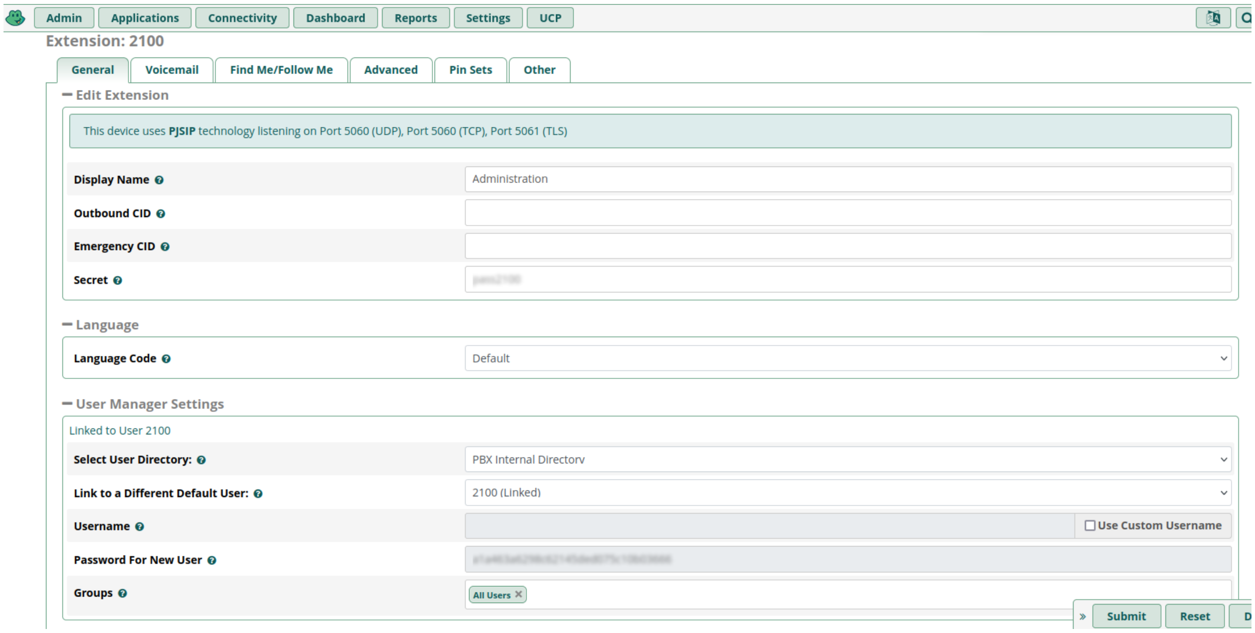
Extension	User Role	Device
2000	Reception 	Zoiper 5 (Windows)
2100	Administration/Admin	Linphone (Windows)

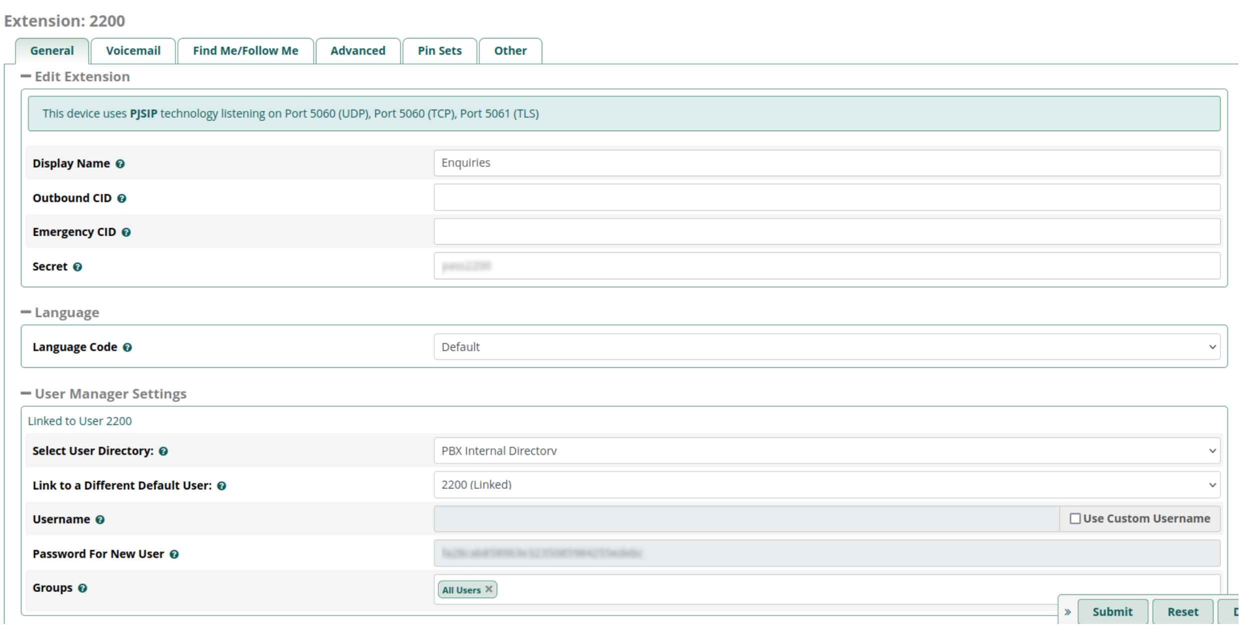
	A General Details Edit your account information. A Add an image SIP address : sip:2100@10.230.27.176 Display name The name displayed to your contacts during exchanges. Admin International code*	

2200	Enquiries  	Linphone (Android)

2300	Business Support  	Sipnetic (Android)


Admin
Applications
Connectivity
Dashboard
Reports
Settings
UCP
Extension: 2000
General
Voicemail
Find Me/Follow Me
Advanced
Pin Sets
Other
Edit Extension
This device uses PJSIP technology listening on Port 5060 (UDP), Port 5060 (TCP), Port 5061 (TLS)
Display Name
Reception
Outbound CID
Emergency CID
Secret
Language
Language Code
Default
User Manager Settings
Linked to User 2000
Select User Directory:
PBX Internal Directory
Link to a Different Default User:
2000 (Linked)
Username
Use Custom Username
Password For New User
Groups
All Users
Submit
Reset


Admin
Applications
Connectivity
Dashboard
Reports
Settings
UCP
Extension: 2100
General
Voicemail
Find Me/Follow Me
Advanced
Pin Sets
Other
Edit Extension
This device uses PJSIP technology listening on Port 5060 (UDP), Port 5060 (TCP), Port 5061 (TLS)
Display Name
Administration
Outbound CID
Emergency CID
Secret
Language
Language Code
Default
User Manager Settings
Linked to User 2100
Select User Directory:
PBX Internal Directory
Link to a Different Default User:
2100 (Linked)
Username
Use Custom Username
Password For New User
Groups
All Users
Submit
Reset


Extension: 2200
General
Voicemail
Find Me/Follow Me
Advanced
Pin Sets
Other
Edit Extension
This device uses PJSIP technology listening on Port 5060 (UDP), Port 5060 (TCP), Port 5061 (TLS)
Display Name
Enquiries
Outbound CID
Emergency CID
Secret
Language
Language Code
Default
User Manager Settings
Linked to User 2200
Select User Directory:
PBX Internal Directory
Link to a Different Default User:
2200 (Linked)
Username
Use Custom Username
Password For New User
Groups
All Users
Submit
Reset

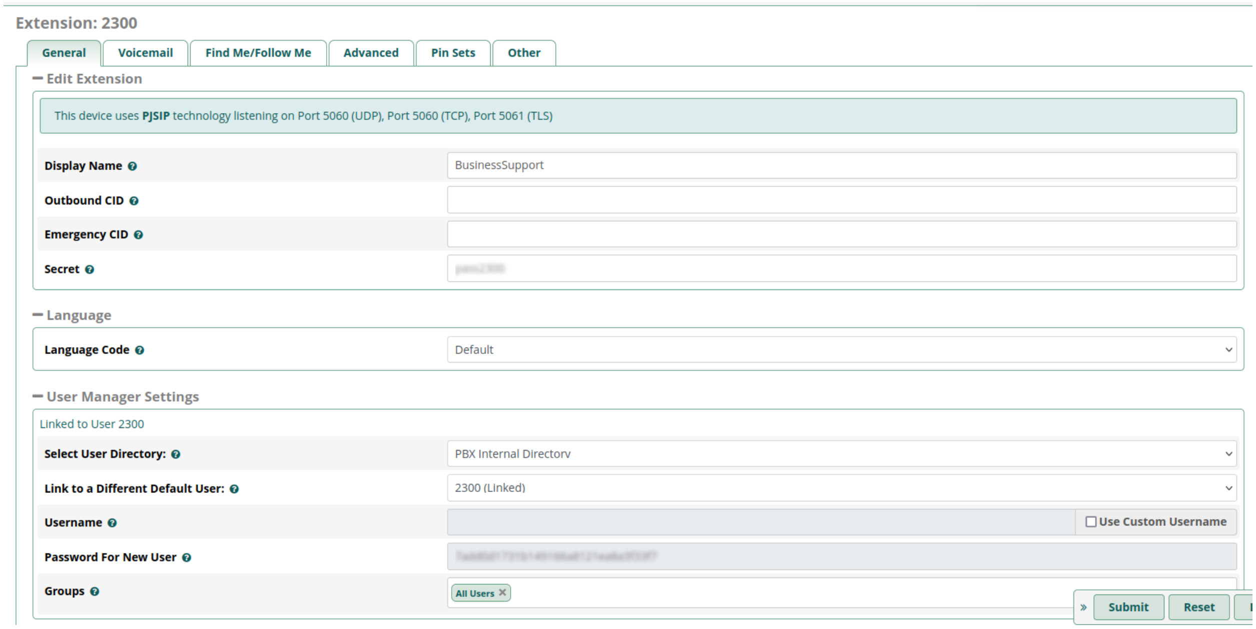


Figure 2: SIP extension configuration within the FreePBX PBX environment

4.2 SIP Registration Testing

All configured SIP extensions were successfully registered with the FreePBX server. Internal VoIP connectivity between desktop and mobile SIP clients was validated satisfactorily.

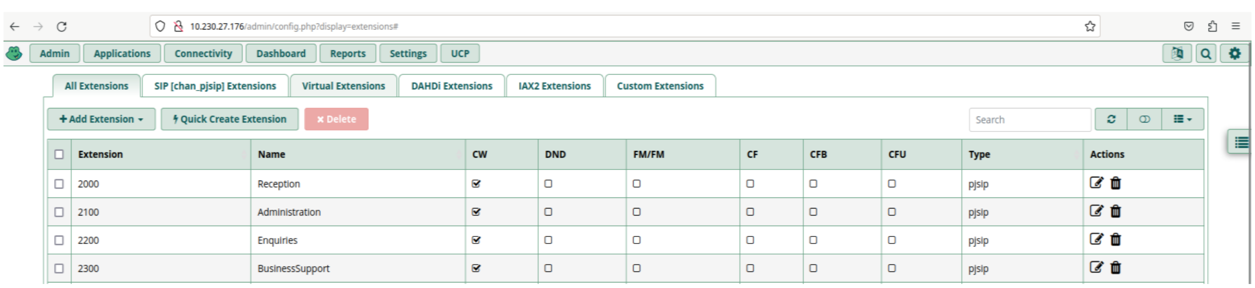
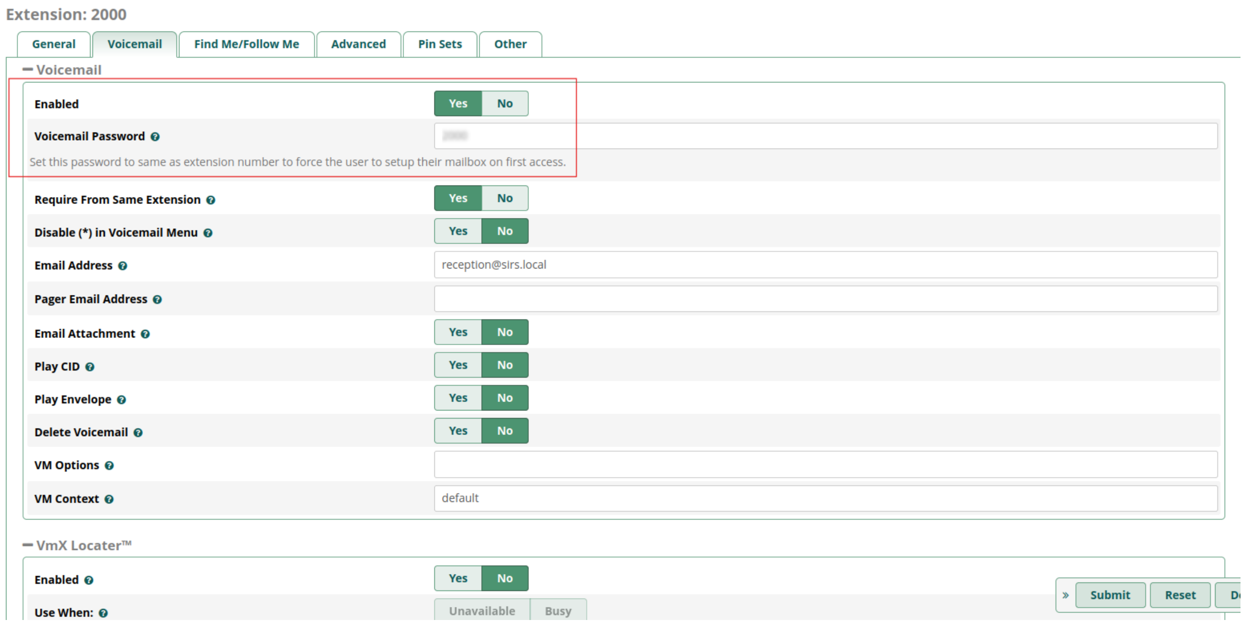
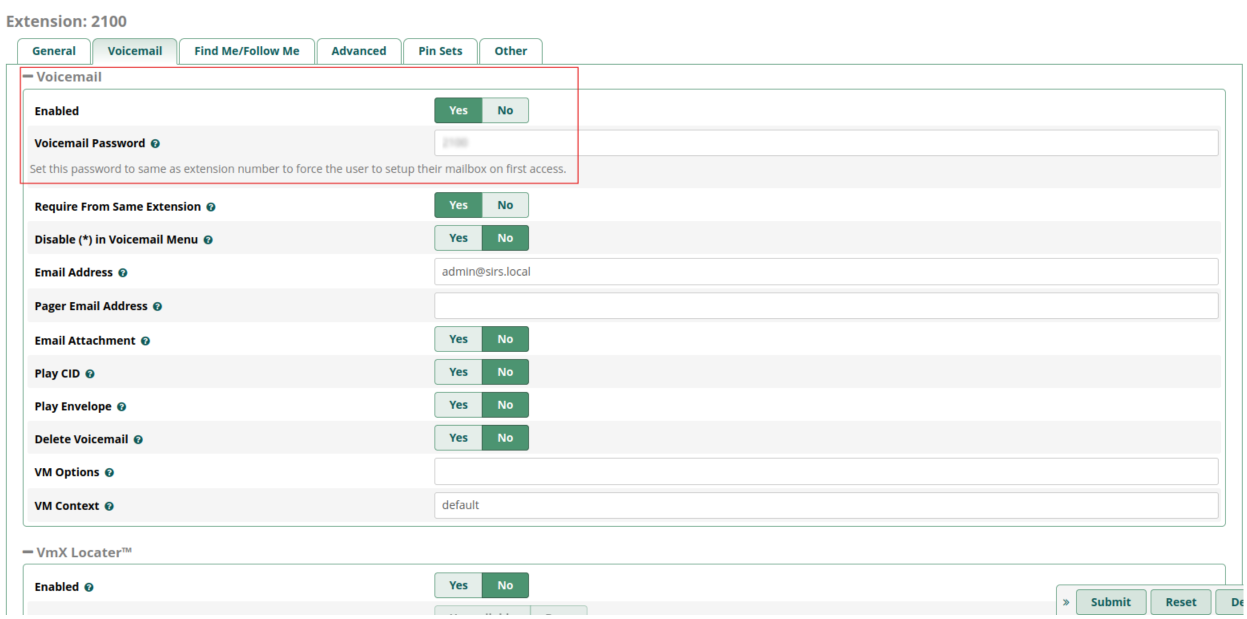


Figure 3: Successful SIP extension registration within the PBX system.

4.3 Voicemail Configuration

Voicemail functionality was enabled for all configured extensions to facilitate business continuity and distant accessibility.





4.3.1 Voicemail Access Testing

The voicemail retrieval feature was successfully tested using SIP softphones. The voicemail system was accessed internally by calling the FreePBX voicemail feature code.

Feature Code	Purpose
*97	Direct voicemail access

Testing was accomplished using the Sipnetic Android SIP client to validate effective voicemail authentication and message retrieval.

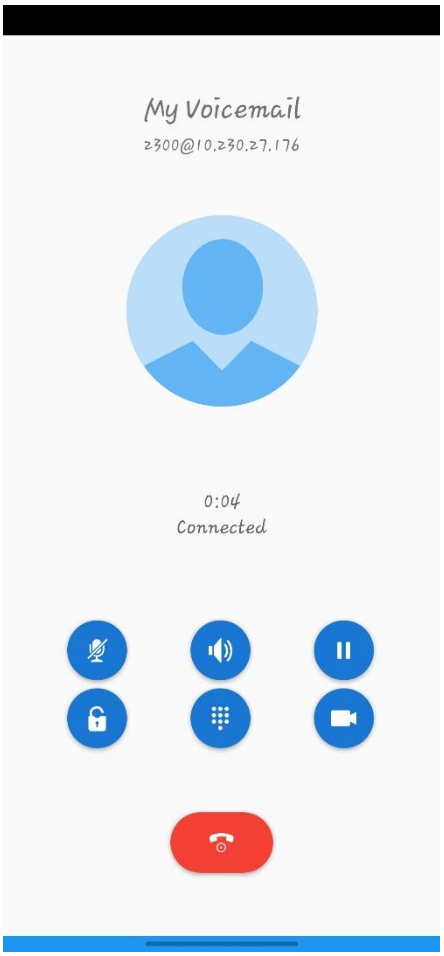


Figure 4: Successful voicemail retrieval with the *97 voicemail feature code.

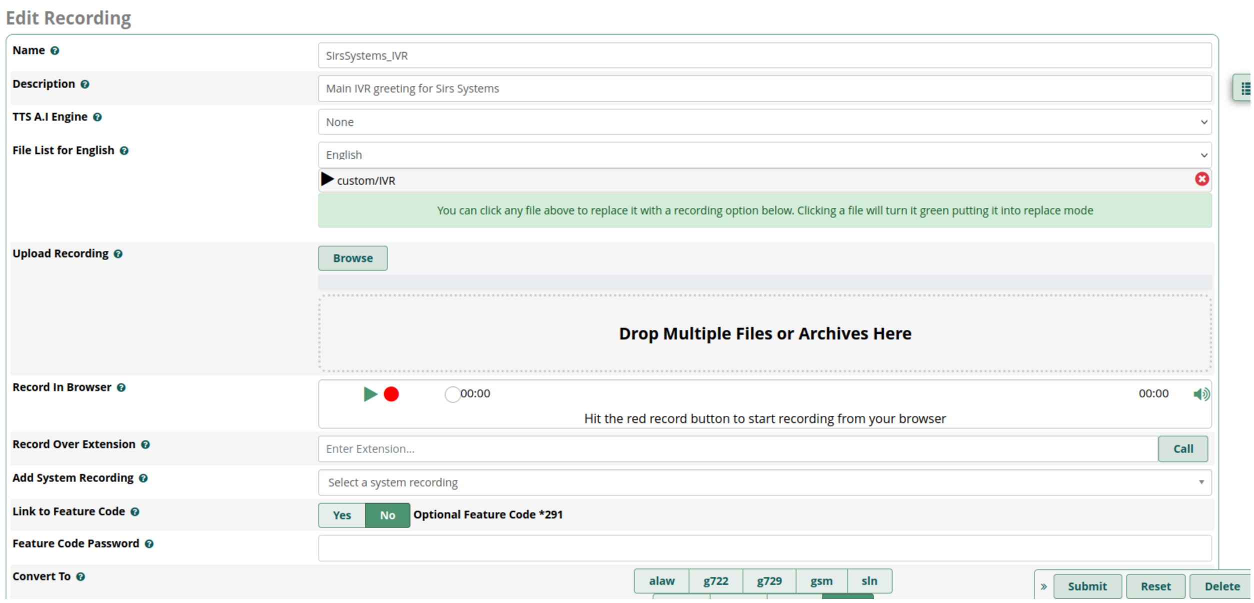
The voicemail system allowed users to access stored messages remotely, boosting communication dependability and ensuring missed calls could be examined efficiently inside the VoIP context.

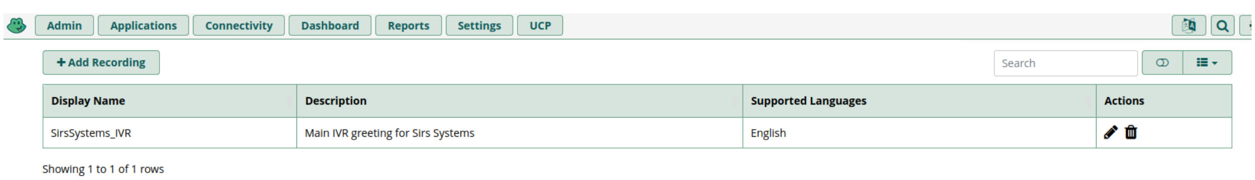
4.4 IVR / Auto-Attendant Configuration

An IVR system was setup to provide automatic call routing features for incoming callers.

The IVR menu included:

Menu Option	Destination
Press 1	Reception
Press 2	Administration
Press 3	Enquiries
Press 4	Business Support
Press 5	Conference Room





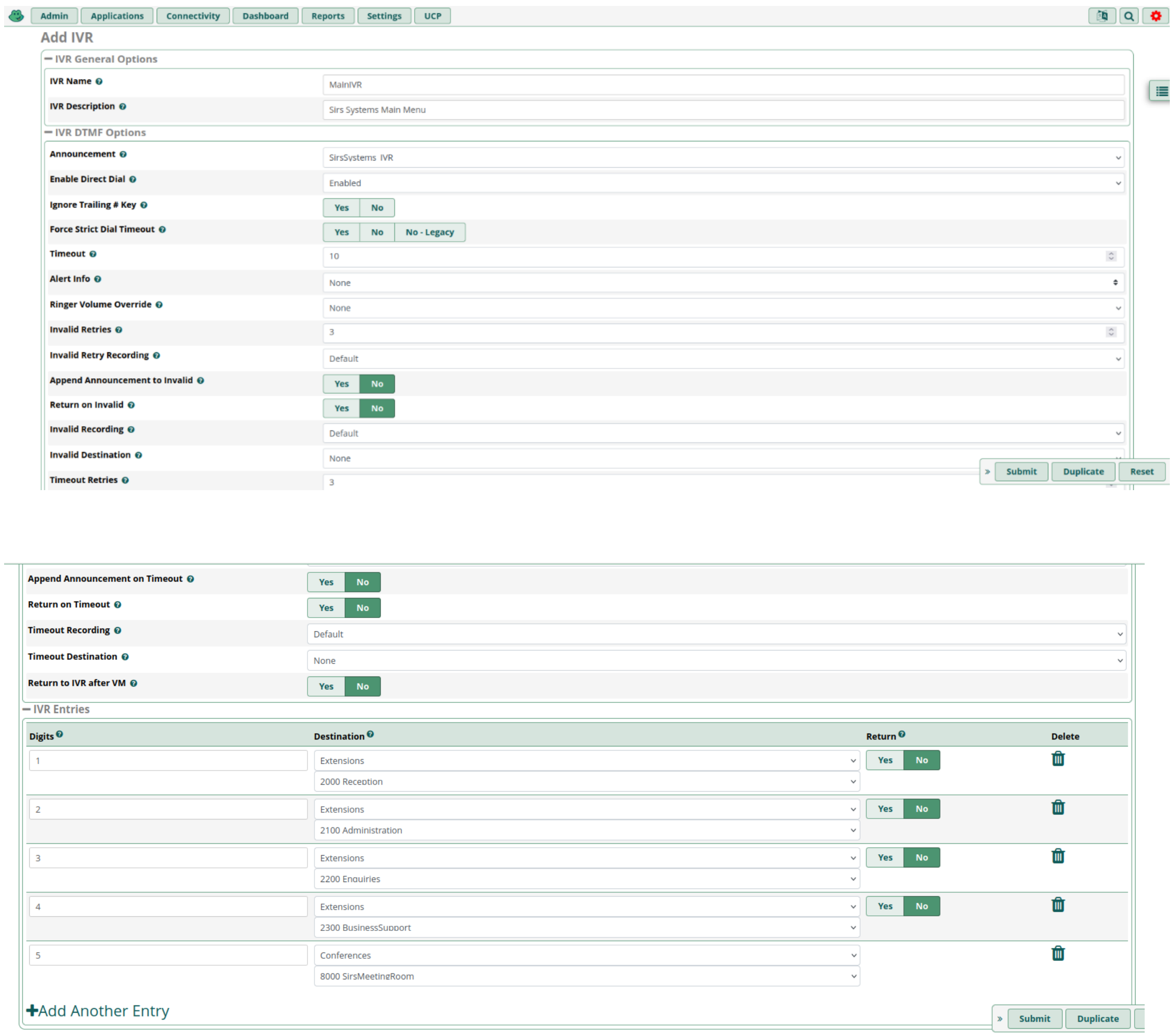
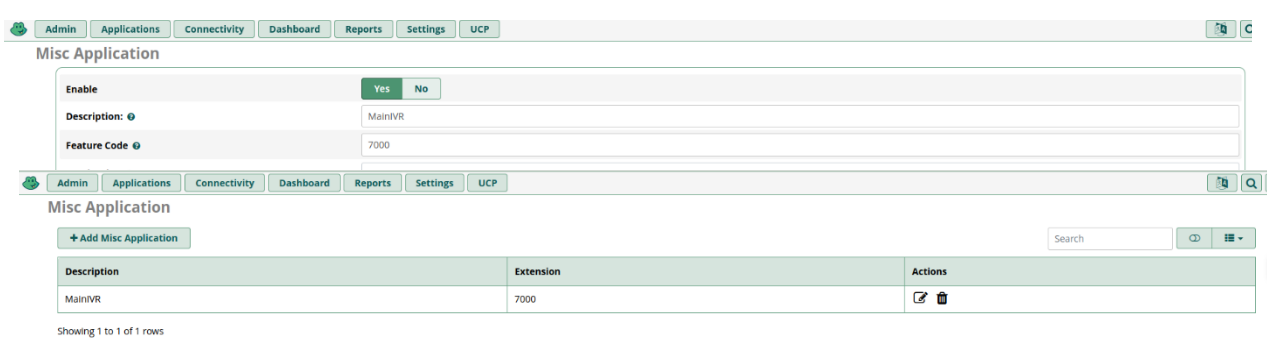


Figure 5: IVR auto-attendant configuration within FreePBX.

Create Internal IVR Extension (7000)



4.5 IVR Testing

The IVR system was tested successfully utilising internal SIP extensions. Call routing functionality performed correctly for various extension destinations.

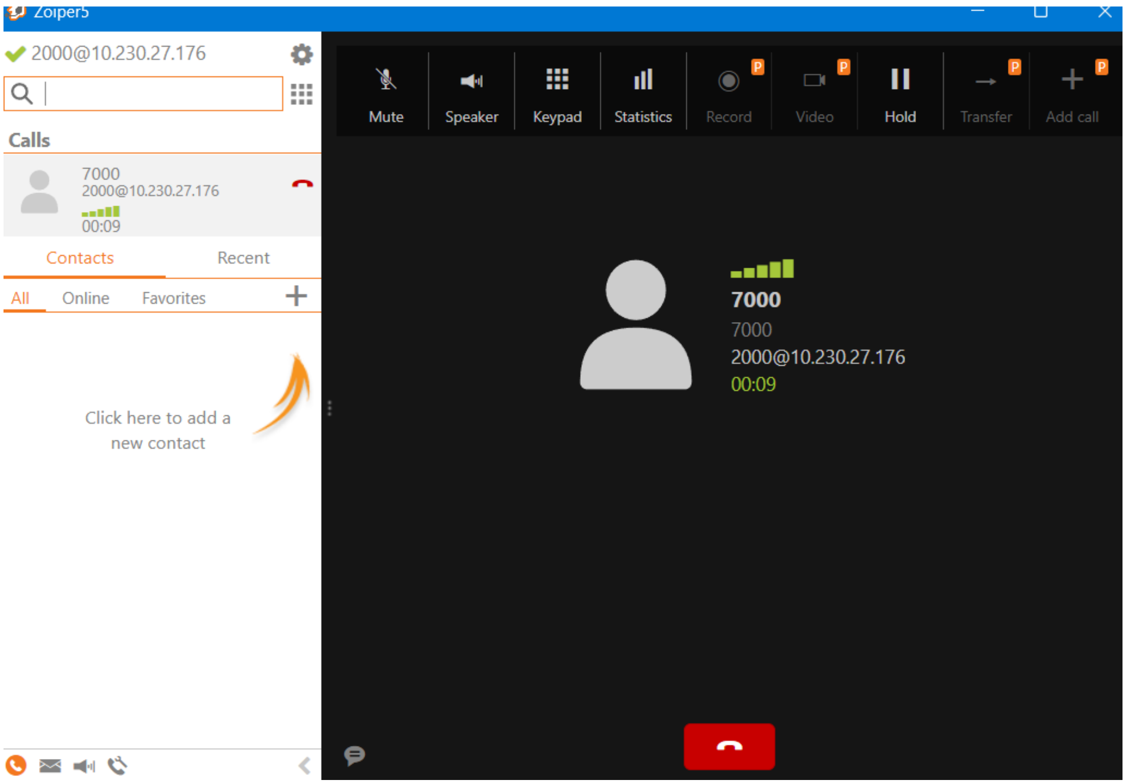


Figure 6: Successful IVR testing and internal call routing demonstration.

5. Level 2 System Implementation

5.1 Ring Group Configuration

Ring Groups were established to support geographically scattered workers functioning remotely across various sites.

- Staff Ring Group

Ring Group	Extensions
6000	2000, 2200

- Support Ring Group

Ring Group	Extensions
6100	2100, 2300

The ringall technique was adopted to facilitate simultaneous alerting across different SIP endpoints.

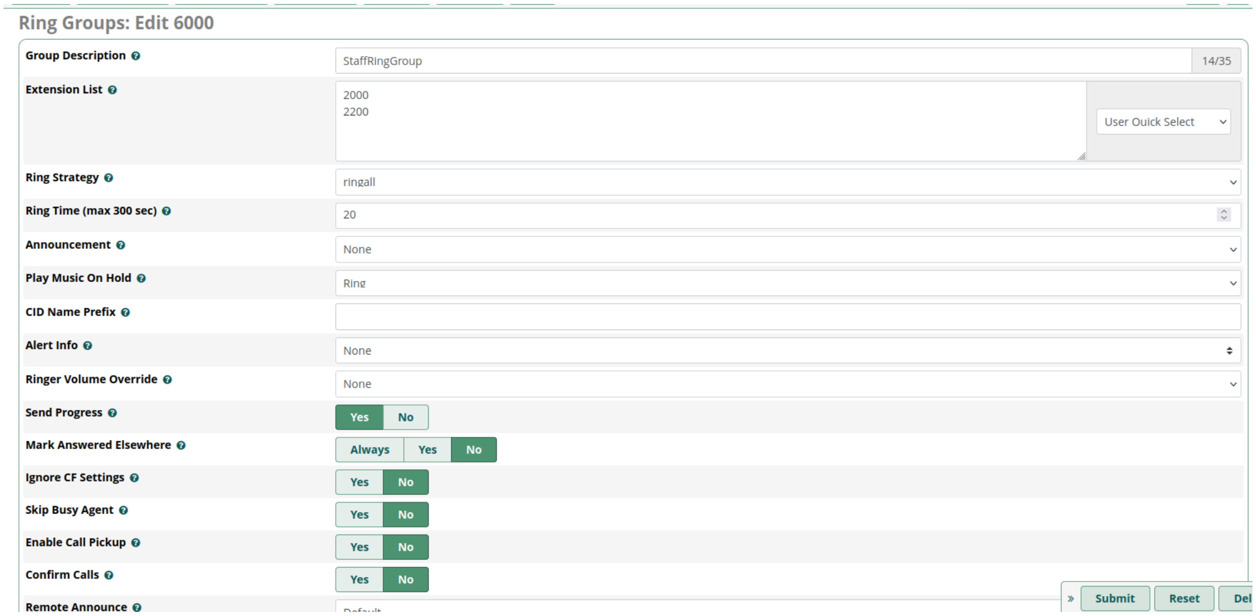


Figure 7: Ring Group configuration supporting simultaneous SIP endpoint ringing.

5.2 Ring Group Testing

Testing demonstrated that both desktop and mobile SIP clients rang simultaneously when Ring Group extensions were dialled.

The Administration extension successfully launched calls to both the Reception and Enquiries extensions using the set Ring Group (6000) functionality.

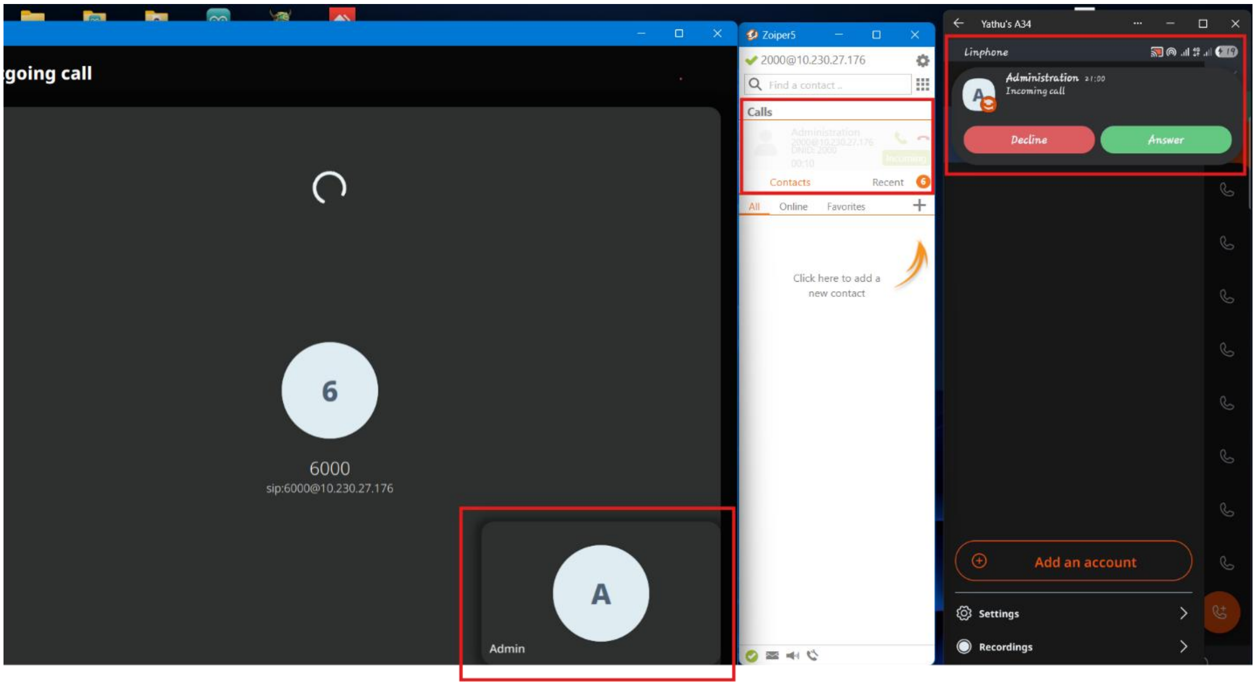


Figure 8: Successful Ring Group operation across several remote SIP devices.

5.3 Call Blocking / Blacklisting

Call blocking feature was built using the FreePBX blacklist module to prevent unwanted or malicious calls from accessing internal SIP extensions.

A test extension was added to the blacklist and call rejection behaviour was validated correctly.

For testing purpose, I block 2200 (Enquires) extension

Figure 9: FreePBX blacklist setting used to block unwanted SIP calls.

6. Level 3 System Implementation

6.1 Audio Conferencing

Audio conferencing capabilities was created using the FreePBX Conferences module.

Conference Room:

Conference Number	Purpose
8000	SirsMeetingRoom

The conference function enabled several geographically scattered users to converse simultaneously through the PBX environment.

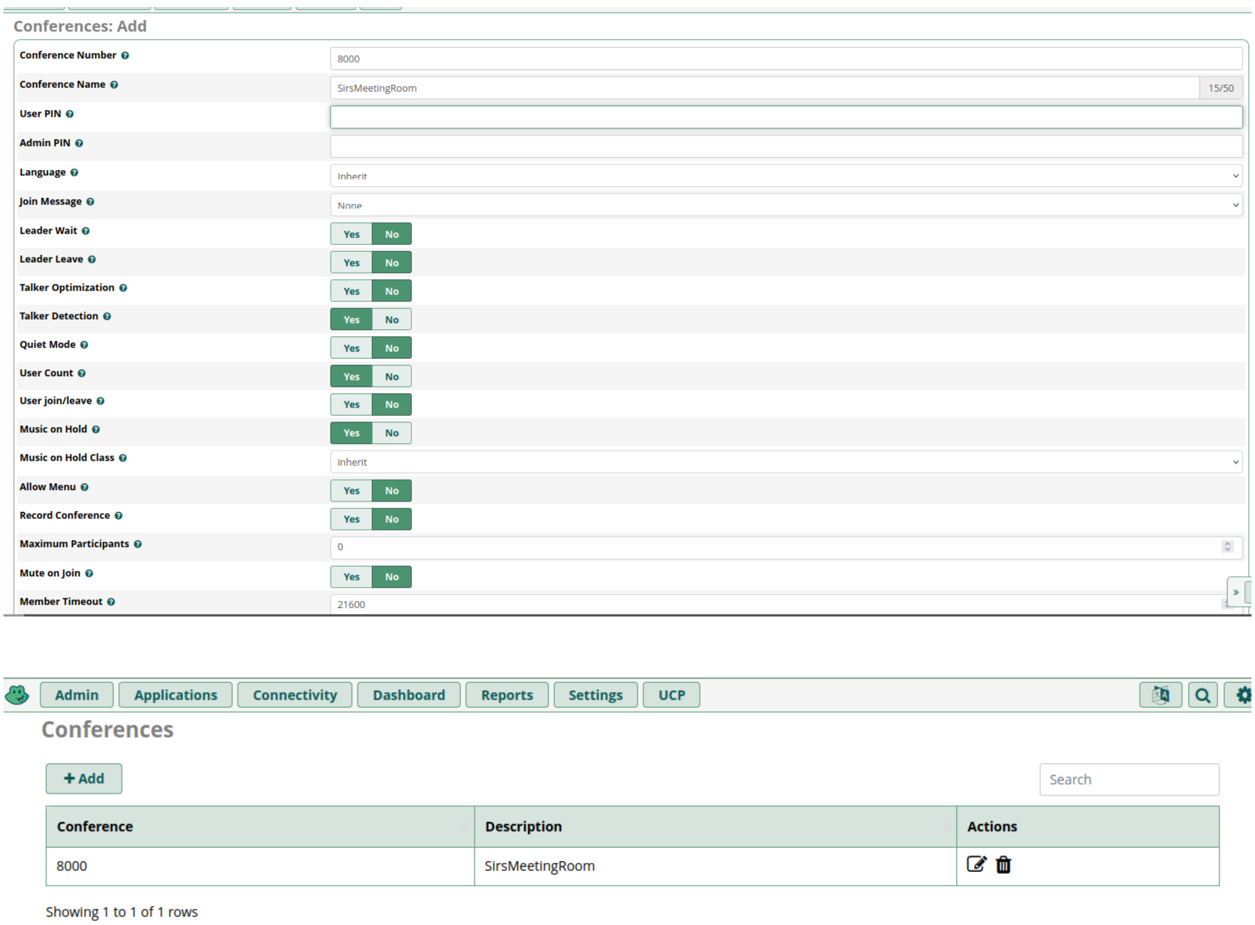
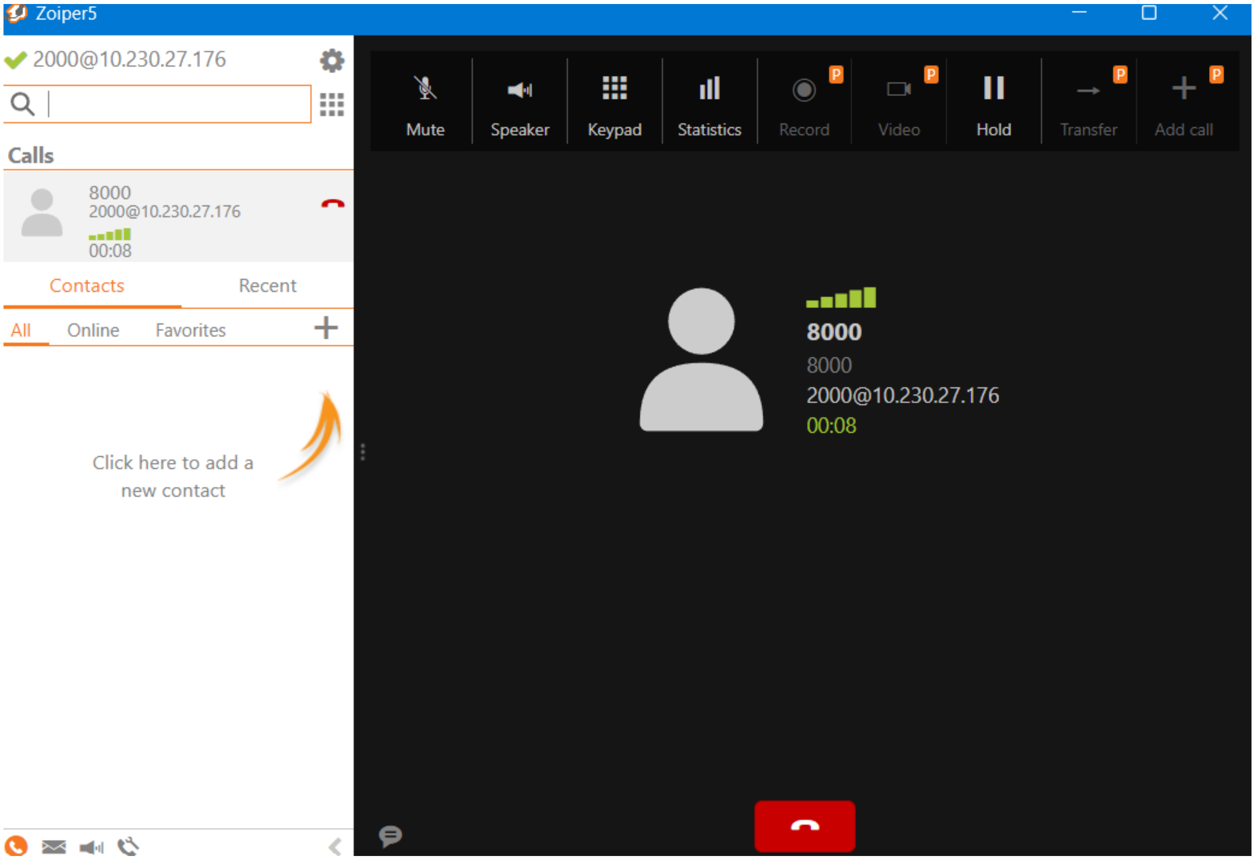


Figure 10: Conference room setting within the FreePBX PBX system.

6.2 Conference Testing

Conference functionality was successfully tested utilising desktop and mobile SIP softphones.



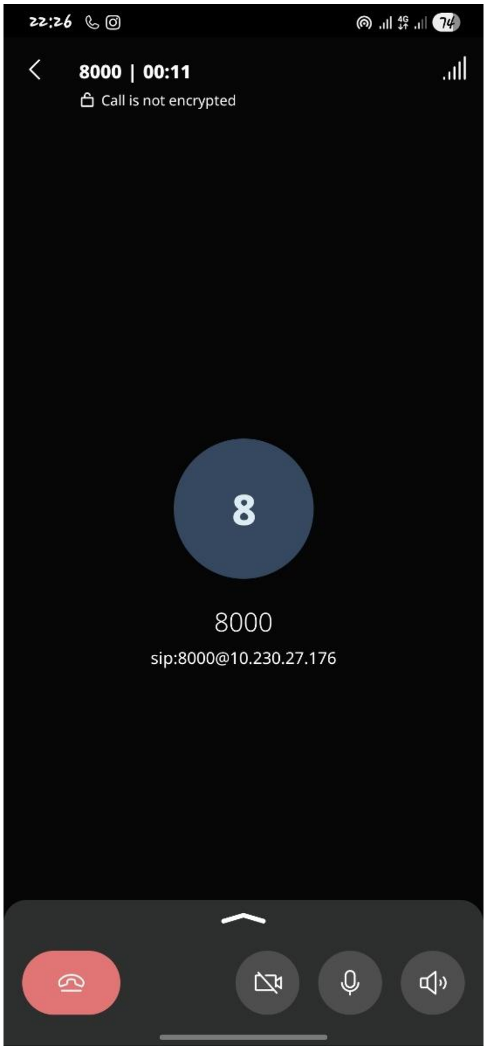


Figure 11: Active VoIP conference session between numerous SIP endpoints

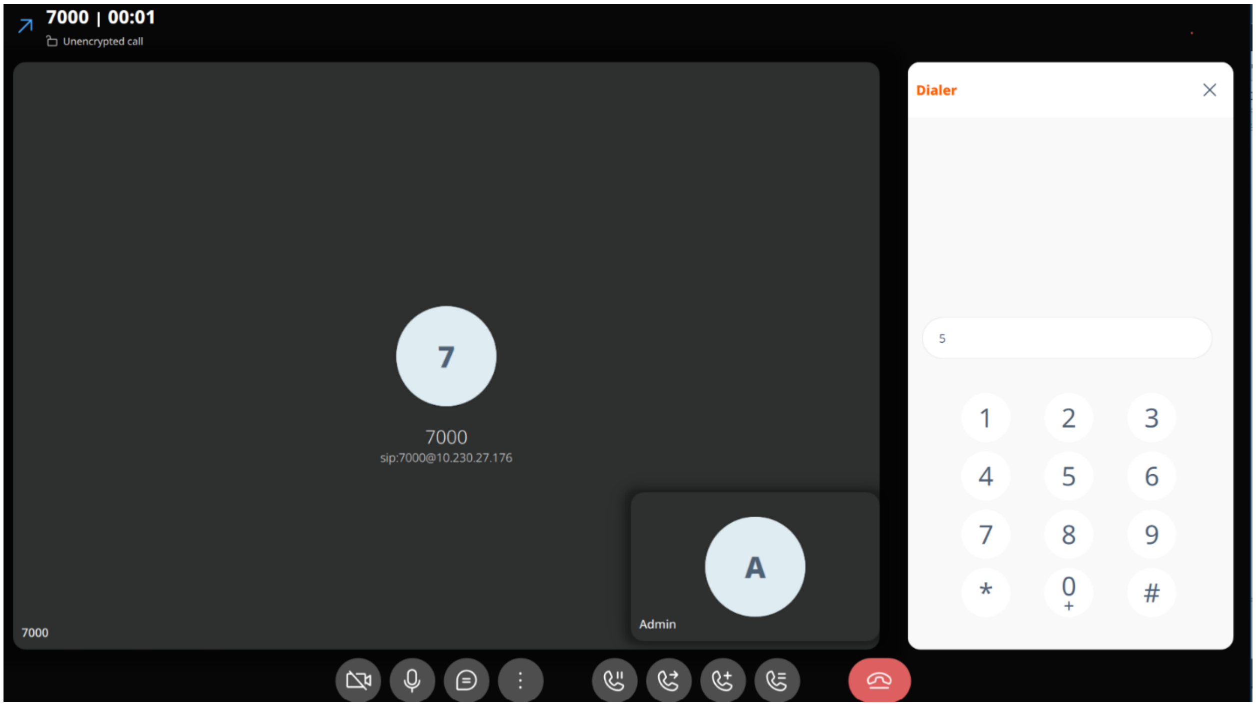
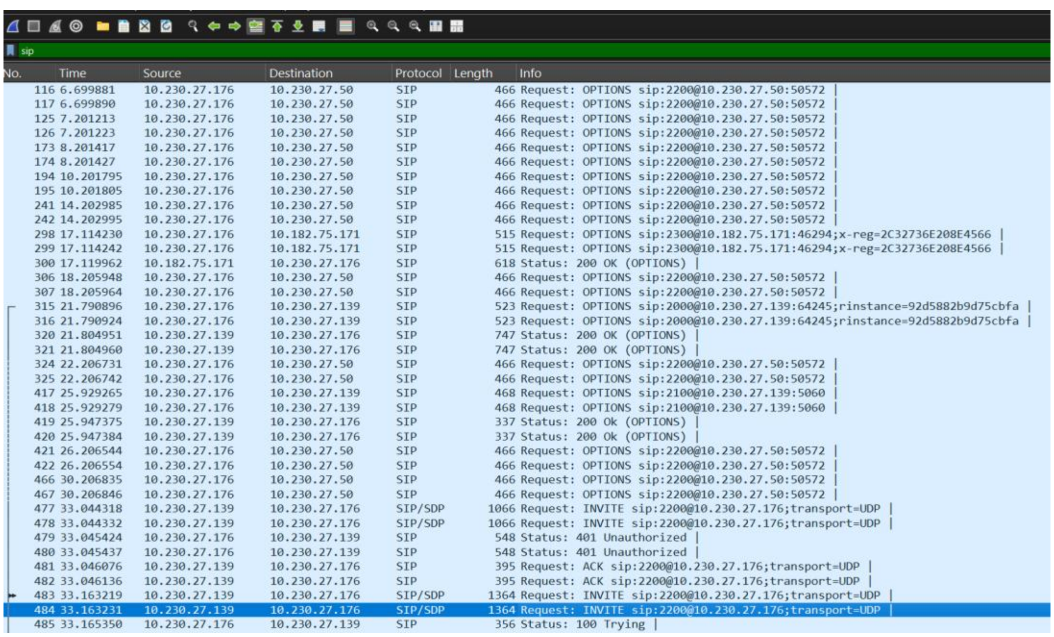


Figure 12: IVR menu integrated with the VoIP conference room to support remote collaborative communication.

6.3 VoIP Traffic Capture and Eavesdropping Demonstration

Wireshark packet analysis proved that normal SIP signalling communication can be viewed across the network when transmitted without encryption.



No.	Time	Source	Destination	Protocol	Length	Info
116	6.699881	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
117	6.699890	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
125	7.201213	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
126	7.201223	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
173	8.201417	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
174	8.201427	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
194	10.201795	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
195	10.201805	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
241	14.202985	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
242	14.202995	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
298	17.114230	10.230.27.176	10.182.75.171	SIP	515	Request: OPTIONS sip:2300@10.182.75.171:46294;x-reg=2C32736E208E4566 |
299	17.114242	10.230.27.176	10.182.75.171	SIP	515	Request: OPTIONS sip:2300@10.182.75.171:46294;x-reg=2C32736E208E4566 |
300	17.119962	10.182.75.171	10.230.27.176	SIP	618	Status: 200 OK (OPTIONS) |
306	18.205948	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
307	18.205964	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
315	21.790896	10.230.27.176	10.230.27.139	SIP	523	Request: OPTIONS sip:2000@10.230.27.139:64245;rinstance=92d5882b9d75cbfa |
316	21.790924	10.230.27.176	10.230.27.139	SIP	523	Request: OPTIONS sip:2000@10.230.27.139:64245;rinstance=92d5882b9d75cbfa |
320	21.804951	10.230.27.139	10.230.27.176	SIP	747	Status: 200 OK (OPTIONS) |
321	21.804960	10.230.27.139	10.230.27.176	SIP	747	Status: 200 OK (OPTIONS) |
324	22.206731	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
325	22.206742	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
417	25.929265	10.230.27.176	10.230.27.139	SIP	468	Request: OPTIONS sip:2100@10.230.27.139:5060 |
418	25.929279	10.230.27.176	10.230.27.139	SIP	468	Request: OPTIONS sip:2100@10.230.27.139:5060 |
419	25.947375	10.230.27.139	10.230.27.176	SIP	337	Status: 200 Ok (OPTIONS) |
420	25.947384	10.230.27.139	10.230.27.176	SIP	337	Status: 200 Ok (OPTIONS) |
421	26.206544	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
422	26.206554	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
466	30.206835	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
467	30.206846	10.230.27.176	10.230.27.50	SIP	466	Request: OPTIONS sip:2200@10.230.27.50:50572 |
477	33.044318	10.230.27.139	10.230.27.176	SIP/SDP	1066	Request: INVITE sip:2200@10.230.27.176;transport=UDP |
478	33.044332	10.230.27.139	10.230.27.176	SIP/SDP	1066	Request: INVITE sip:2200@10.230.27.176;transport=UDP |
479	33.045424	10.230.27.176	10.230.27.139	SIP	548	Status: 401 Unauthorized |
480	33.045437	10.230.27.176	10.230.27.139	SIP	548	Status: 401 Unauthorized |
481	33.046076	10.230.27.139	10.230.27.176	SIP	395	Request: ACK sip:2200@10.230.27.176;transport=UDP |
482	33.046136	10.230.27.139	10.230.27.176	SIP	395	Request: ACK sip:2200@10.230.27.176;transport=UDP |
483	33.163219	10.230.27.139	10.230.27.176	SIP/SDP	1364	Request: INVITE sip:2200@10.230.27.176;transport=UDP |
484	33.163231	10.230.27.139	10.230.27.176	SIP/SDP	1364	Request: INVITE sip:2200@10.230.27.176;transport=UDP |
485	33.165350	10.230.27.176	10.230.27.139	SIP	356	Status: 100 Trying |

During testing, SIP INVITE and REGISTER messages exposed endpoint addressing information, extension numbers and session negotiation details.

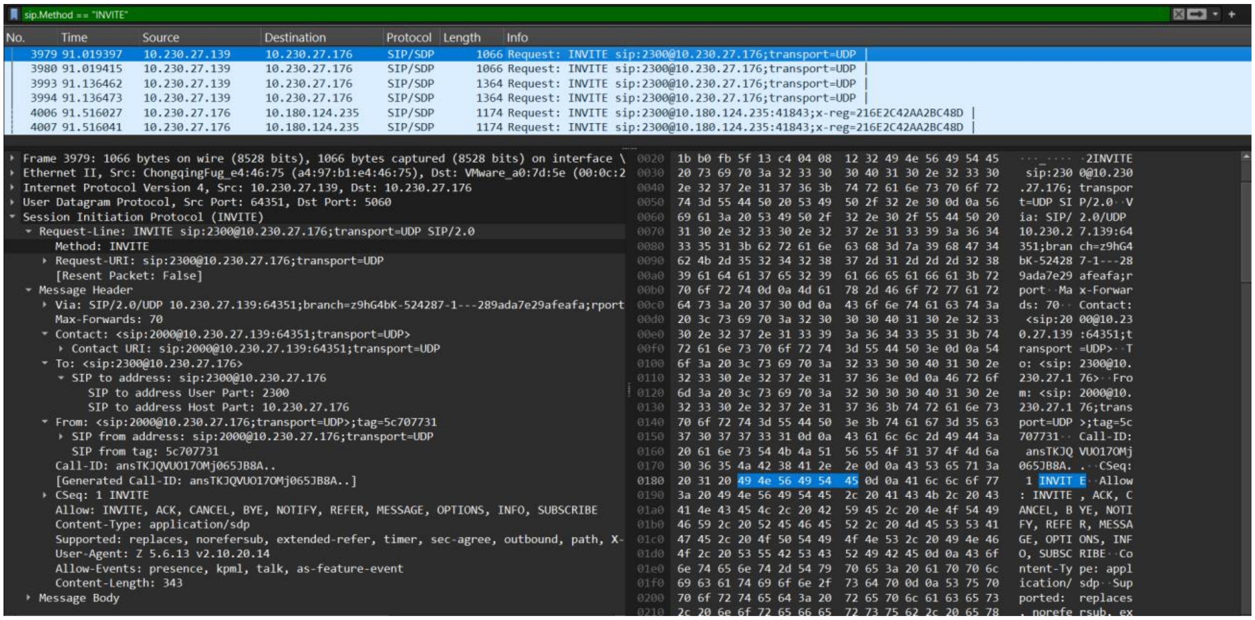


Figure 13: SIP signalling traffic obtained using Wireshark.

6.4 VoIP Security Configuration

VoIP security measures were studied within the FreePBX PBX environment through the implementation of TLS-based SIP transport and SRTP media encryption settings. Although full end-to-end encrypted communication was not regularly achieved due to interoperability issues between SIP softphone applications and self-signed certificates, the implementation effectively demonstrated the principles of secure VoIP communication and SIP transport security.

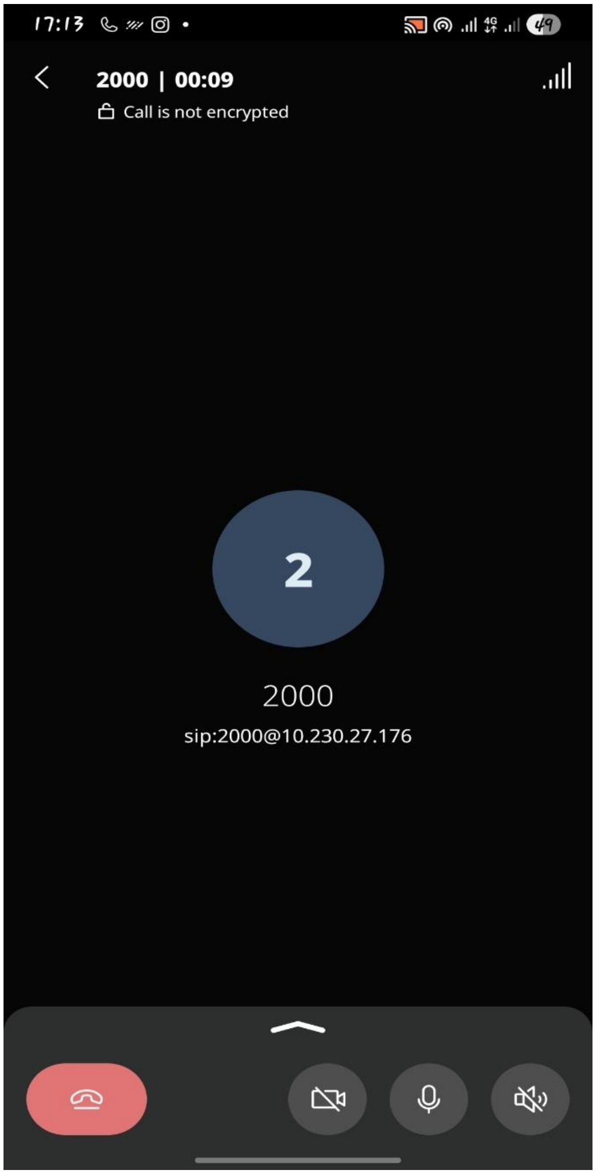


Figure 14: During testing, the softphone applications indicated that active calls remained unencrypted, underscoring the significance of secure VoIP protocols such as TLS and SRTP for protecting voice communication from interception and eavesdropping assaults.

6.5 Firewall Configuration

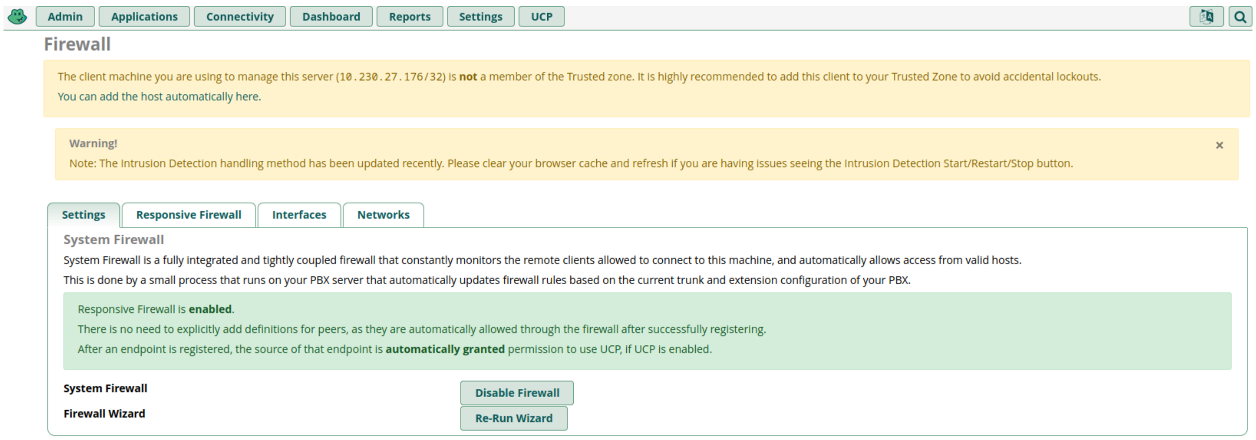


Figure 15: FreePBX firewall setup used to increase PBX security and network access management.

The built-in FreePBX Firewall module was enabled to increase the overall security of the PBX environment. The firewall provides extra protection by preventing unauthorised access attempts to SIP services and PBX administration interfaces. Trusted local network zones were permitted but external access remained regulated by the FreePBX security settings. This additional security layer helps lower the risk of hostile VoIP attacks and unauthorised system access.

7. Additional Features

7.1 Music on Hold

An additional Music on Hold (MOH) capability was integrated into the FreePBX PBX system to enhance the caller experience during wait and transfer procedures.

The default MOH class supplied by FreePBX was employed and successfully tested in Ring Group and call handling scenarios.

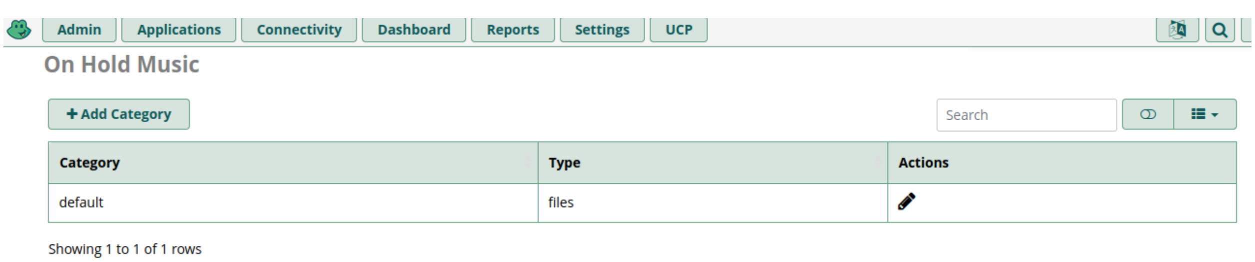


Figure 16: Default Music on Hold settings are accessible within the FreePBX PBX system.

8. Testing and Results

Feature	Tested	Result
SIP Registration	Yes	Successful
Internal Calling	Yes	Successful
Voicemail	Yes	Successful
IVR	Yes	Successful
Ring Groups	Yes	Successful
Conferencing	Yes	Successful
Blacklisting	Yes	Successful
Wireshark Packet Capture	Yes	Successful
Music on Hold	Yes	Successful

9. Conclusion

The practical implementation successfully demonstrated the deployment and configuration of a complete VoIP PBX environment utilising FreePBX and Asterisk technologies. The completed system featured basic business communication features including SIP extensions, voicemail, IVR call routing, Ring Groups, conferencing facilities, call blocking and VoIP security configuration.

The project proved how VoIP technology may serve geographically spread businesses through flexible and cost-effective communication solutions running across desktop and mobile platforms. In addition, the research studied VoIP security topics including SIP traffic analysis, secure transport configuration and the relevance of encrypted communication protocols.

Overall, the entire solution fulfilled the objectives of the Sirs Systems scenario while exhibiting practical mastery of current VoIP deployment, configuration and security principles inside a virtualised network environment.